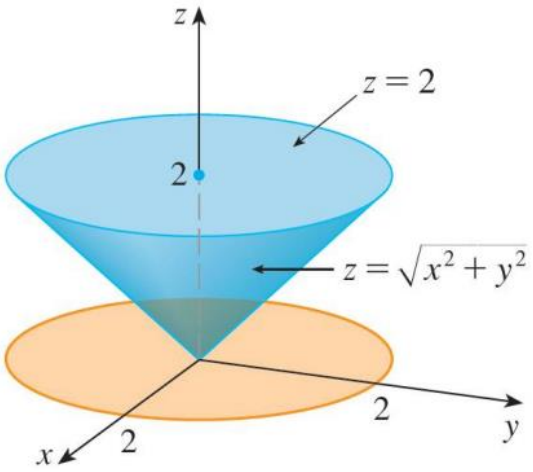
Vector Calculus and Complex Analysis Homework 6

1. Consider the curved surface S of a cone in the below figure. Verify Stokes's Theorem for $\boldsymbol{F}(x,y,z)=\langle xz,2xy,3yz\rangle$.



Solution

The parameterization of the curved surface S is

$$\boldsymbol{r}(u,v)=(u\cos v,u\sin v,u),\quad 0\le u\le 2,\ 0\le v\le 2\pi$$

The upward normal vector is $\boldsymbol{r}_u\times\boldsymbol{r}_v=(\cos v,\sin v,1)\times(-u\sin v,u\cos v,0)=(-u\cos v,-u\sin v,u)$

$C:(2\cos s,2\sin s,2)\quad 0\le s\le 2\pi.$

$\boldsymbol{F}(x,y,z)=\langle xz,2xy,3yz\rangle.$

$\text{curl}(\boldsymbol{F})=(3z,x,2y)$

$\boldsymbol{r}(u,v)=(u\cos v,u\sin v,u),\quad 0\le u\le 2,\ 0\le v\le 2\pi$

$\boldsymbol{r}_u\times\boldsymbol{r}_v=(\cos v,\sin v,1)\times(-u\sin v,u\cos v,0)=(-u\cos v,-u\sin v,u)$

$\iint_S \text{curl}(\boldsymbol{F})\cdot\boldsymbol{n}dA=\iint_R(3u,u\cos v,2u\sin v)\cdot(-u\cos v,-u\sin v,u)dudv$

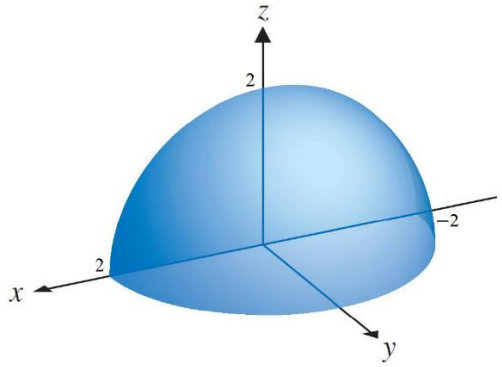
$=\int_0^{2\pi}\int_0^2 u^2(-3\cos v-2\sin v\cos v+2\sin v)dudv$

$=\frac{8}{3}\int_0^{2\pi}(-3\cos v-2\sin v\cos v+2\sin v)dv$

$=\frac{8}{3}\left[-3\sin v-2\sin^2 v-2\cos v\right]_0^{2\pi}=0$

$\int_C \boldsymbol{F}\cdot r'(u)du=4\int_0^{2\pi}(\cos s,2\sin s\cos s,3\sin s)\cdot(-2\sin s,2\cos s,0)ds=\left[\text{function of sin and cos}\right]_0^{2\pi}=0$

2. Consider the surface S satisfied $y, z \geq 0, x^2 + y^2 + z^2 = 4$. Verify Stokes's Theorem for $\boldsymbol{F}(x, y, z) = \langle yz, 2xz, 3xy \rangle$.



Solution

$\boldsymbol{r}(u, v) = (2\cos v \cos u, 2\cos v \sin u, 2\sin v) \quad 0 \leq u \leq \pi, \quad 0 \leq v \leq \frac{\pi}{2}.$

$C_1 = (2\cos u, 2\sin u, 0) \quad 0 \leq u \leq \pi$

$C_2 = (2\cos u, 0, 2\sin u) \quad 0 \leq u \leq \pi$

$\boldsymbol{r}_u = (-2\cos v \sin u, 2\cos v \cos u, 0)$

$\boldsymbol{r}_v = (-2\sin v \cos u, -2\sin v \sin u, 2\cos v)$

$N = 2\cos v(2\cos v \cos u, 2\cos v \sin u, 2\sin v)$

$C = C_1 - C_2$

$\boldsymbol{F}(x, y, z) = \langle yz, 2xz, 3xy \rangle.$

$\text{curl}(\boldsymbol{F}) = (3x - 2x, y - 3y, 2z - z) = (x, -2y, z)$

$$\iint_S \text{curl}(\boldsymbol{F}) \cdot \boldsymbol{n} dA = \iint_R 2\cos v(2\cos v \cos u, -4\cos v \sin u, 2\sin v) \cdot (2\cos v \cos u, 2\cos v \sin u, 2\sin v) du dv$$

$$= \int_0^{\frac{\pi}{2}} \int_0^{\pi} 2\cos v(4\cos^2 v(\cos^2 u - 2\sin^2 u) + 4\sin^2 v) du dv$$

$$= \int_0^{\frac{\pi}{2}} 2\cos v(4\cos^2 v \left[\frac{u}{2} + \frac{\sin 2u}{4} - 2\left(\frac{u}{2} - \frac{\sin 2u}{4} \right) \right]_0^{\pi} + 4\pi \sin^2 v) dv$$

$$= 2\int_0^{\frac{\pi}{2}} \cos v(-2\pi(1 - \sin^2 v) + 4\pi \sin^2 v) dv$$

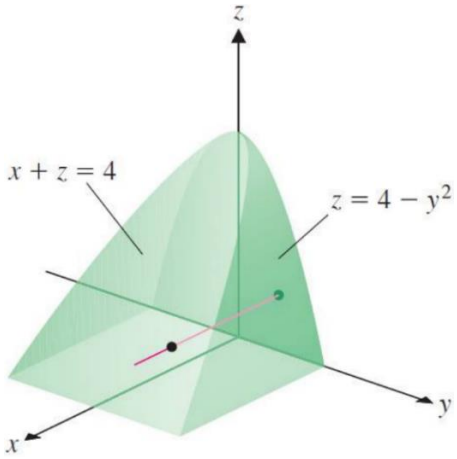
$$= 2\int_0^{\frac{\pi}{2}} (-2\pi(1 - \sin^2 v) + 4\pi \sin^2 v) d\sin v$$

$$= 2\left[-2\pi \left(\sin v - \frac{\sin^3 v}{3} \right) + \frac{4\pi \sin^3 v}{3} \right]_0^{\frac{\pi}{2}}$$

$$= 2\left[-\frac{4\pi}{3} + \frac{4\pi}{3} \right] = 0$$

$$\int_C \boldsymbol{F} \cdot r'(u) du = \int_0^{\pi} 12\cos u \sin u \cdot 0 du - \int_0^{\pi} 8\cos u \sin u \cdot 0 du = 0$$

3. Consider the curved surface S defined by $z=4-y^2, x+z\le 4, x, y, z\ge 0$. Verify Stokes's Theorem for $\boldsymbol{F}(x,y,z)=\langle xy, 2yz, 3xz\rangle$.



Solution

The parameterization of the curved surface S is

$$\boldsymbol{r}(u,v)=(u,v,4-v^2),\quad 0\le u\le 4-(4-v^2)=v^2, 0\le v\le 2\}$$

The outward normal vector is $\boldsymbol{r}_u\times\boldsymbol{r}_v=(1,0,0)\times(0,1,-2v)=(0,2v,1)$

$C_1:(0,s,4-s^2)\quad 2\xrightarrow{s}0.$

$C_2:(s^2,s,4-s^2)\quad 0\xrightarrow{s}2.$

$C_3:(s,2,0)\quad 4\xrightarrow{s}0.$

$\boldsymbol{F}(x,y,z)=\langle xy, 2yz, 3xz\rangle.$

$\text{curl}(\boldsymbol{F})=(-2y,-3z,-x)$

$\iint_S \text{curl}(\boldsymbol{F})\cdot\boldsymbol{n}dA=\iint_R(-2v,-3(4-v^2),-u)\cdot(0,2v,1)dudv$

$=-\int_0^2\int_0^{v^2}\left(6v(4-v^2)+u\right)dudv$

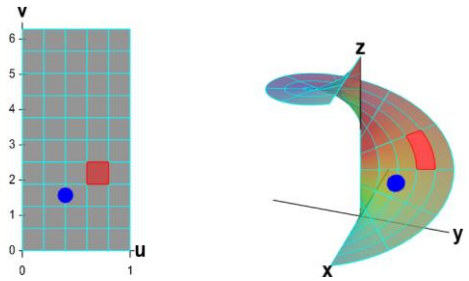
$=-\int_0^2\left(6v^3(4-v^2)+\frac{v^4}{2}\right)dv$

$=-\left[-v^6+\frac{v^5}{10}+6v^4\right]_0^2$

$=-\frac{176}{5}$

$\int_C \boldsymbol{F}\cdot r'(u)du=-\int_0^2 2s(4-s^2)ds+\int_0^2 2s^4+2s(4-s^2)-6s^3(4-s^2)ds-\int_0^4 2sds=-8-\frac{56}{5}-16=-\frac{176}{5}$

4. Consider a helicoid S defined by $r(u,v)=(u\cos v, u\sin v, v)$, $0\le u\le 1, 0\le v\le 2\pi$. Verify Stokes's Theorem for $\boldsymbol{F}(x,y,z)=\langle xy, 2yz, 3xz\rangle$.



Solution

The upward normal vector is $\boldsymbol{r}_u \times \boldsymbol{r}_v = (\cos v, \sin v, 0)\times(-u\sin v, u\cos v, 1) = (\sin v, -\cos v, u)$.

$C_1 : (u,0,0) \quad 0 \xrightarrow{u} 1.$

$C_2 : (\cos v, \sin v, v) \quad 0 \xrightarrow{v} 2\pi.$

$C_3 : (u,0,2\pi) \quad 1 \xrightarrow{u} 0.$

$C_4 : (0,0,v) \quad 2\pi \xrightarrow{v} 0.$

$\boldsymbol{F}(x,y,z)=\langle xy, 2yz, 3xz\rangle.$

$\text{curl}(\boldsymbol{F}) = (-2y, -3z, -x)$

$$\iint_S \text{curl}(\boldsymbol{F})\cdot \boldsymbol{n} dA = \iint_R (-2u\sin v, -3v, -u\cos v)\cdot(\sin v, -\cos v, u) du dv$$

$$= \int_0^{2\pi}\int_0^1 \left(-2u\sin^2 v + 3v\cos v - u^2\cos v\right) du dv$$

$$= \int_0^{2\pi}\left(-\sin^2 v + 3v\cos v - \frac{\cos v}{3}\right) dv$$

$$= -\pi + 3\int_0^{2\pi} v\, d\sin v$$

$$= -\pi + \left[3v\sin v\right]_0^{2\pi} - 3\int_0^{2\pi}\sin v dv = -\pi$$

$$\int_C \boldsymbol{F}\cdot r'(s) ds = \int_0^{2\pi} (\sin v\cos v, 2v\sin v, 3v\cos v)\cdot(-\sin v, \cos v, 1) dv$$

$$= \int_0^{2\pi} (-\sin^2 v\cos v + 2v\sin v\cos v + 3v\cos v) dv$$

$$= \int_0^{2\pi} v\sin 2v dv$$

$$= \int_0^{2\pi} -\frac{v}{2} d\cos 2v$$

$$= -\left[\frac{v}{2}\cos 2v\right]_0^{2\pi} + \int_0^{2\pi}\frac{\cos 2v}{2} dv$$

$$= -\pi$$